

5.3 Managing climate change

Managing the impacts of climate change is a significant undertaking. It requires people around the world to rethink their lifestyles. Managing climate change is contentious because many of the countries that are the least responsible for greenhouse gas emissions are feeling the impacts of climate change the most. There is an argument that the countries that have caused the most damage to the climate should pay the most. This is called **climate justice**. However, dealing with climate change is a global responsibility as everyone will feel the impacts of climate change in the future.

There are two main strategies to managing the impacts of climate change: mitigation approaches and adaptation approaches.

Mitigating the impacts of climate change

Mitigation strategies and techniques aim to reduce or prevent the impacts of climate change.

International agreements and targets

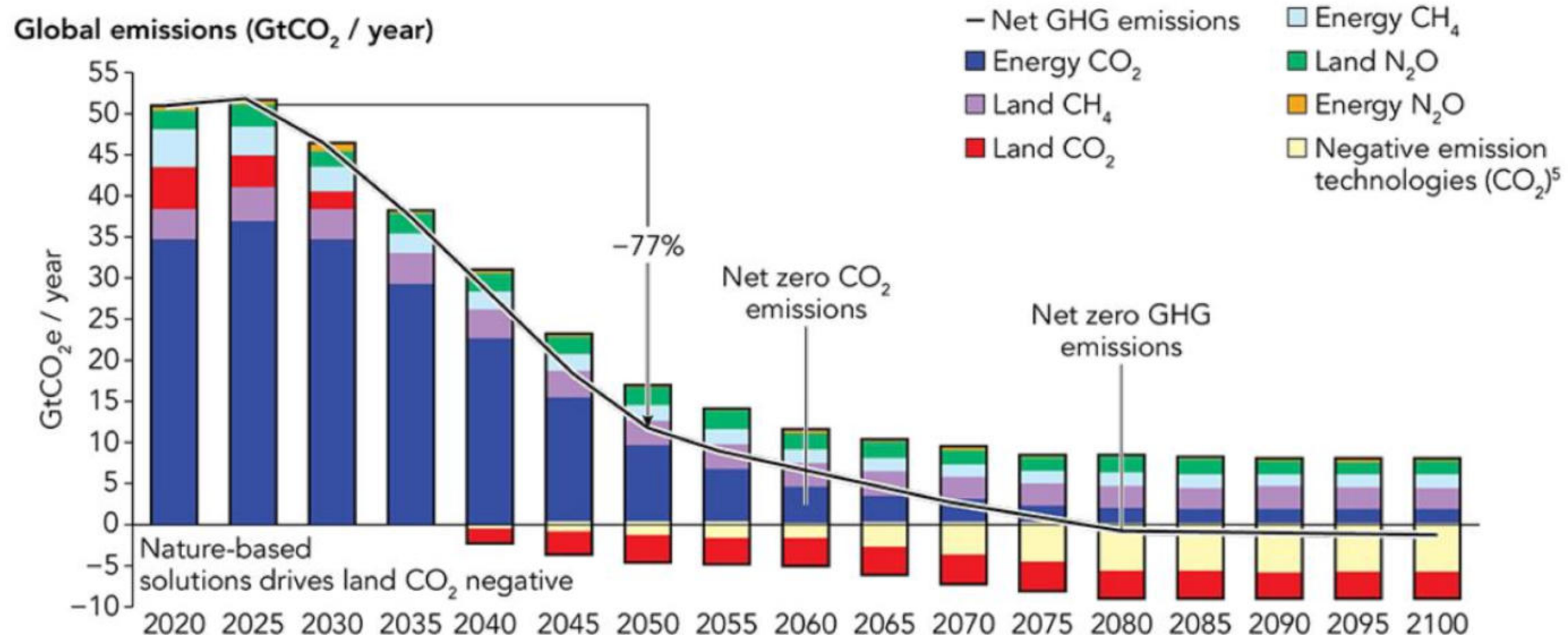
In 1992, 197 countries signed the first global agreement on climate change. It wasn't until 2005, though, that the Kyoto Protocol became international law, with many high-income countries agreeing to reduce their carbon emissions by 5%. Lower-income countries, such as India and Brazil, were not required to make reductions, on the basis of their economic development and their historically smaller contribution to greenhouse gas emissions.

Member states of the United Nations meet every year at COP (Conference of the Parties), a climate change conference, to discuss and agree to initiatives. In 2016, the Paris Agreement united 196 countries at COP21 to keep the increase in global average temperature to well below 2 °C above pre-industrial levels, and to limit global temperature increase to 1.5 °C above pre-industrial levels. Such strategies aim to mitigate the impacts of climate change on an international scale.

National agreements and targets

In recent years, world leaders have stressed the need to limit global warming to 1.5 °C by 2100. Crossing the 1.5 °C threshold risks far more severe climate change impacts. Therefore, most countries set emissions targets in an attempt to limit their greenhouse gas emissions. However, countries often agree to over-ambitious emissions targets, which means they do not keep their emissions pledges.

The USA, for example, has pledged to reduce greenhouse gas emissions to 50–52% below 2005 levels by 2030, achieve 100% carbon-free electricity generation by 2035, and achieve a **net zero** emissions economy by 2050. However, the USA is currently not on track to meet these targets. Such strategies aim to mitigate the impacts of climate change on a national scale.



Source **A**: Global net zero prediction graph

TASK 1: Study Source **A**

- Describe the graph.**
- What year** are we currently expected to reach net zero greenhouse gas emissions?

- c **In 2030, what sector** is predicted to contribute the greatest amount of emissions? What is the contributing greenhouse gas?
- d Explain why low-income countries may be reluctant to reduce their greenhouse emissions.

Waste management

Waste management is a key strategy to mitigate the impacts of climate change. The principle of ‘reduce, reuse, recycle’ promotes more sustainable living:

- ‘Reduce’ refers to minimising the amount of waste we produce. For example, choosing products with minimal packaging or digital alternatives helps cut down on material extraction and manufacturing emissions.
- ‘Reuse’ involves extending the lifespan of products, reducing the need for new items. Embracing practices like using reusable containers, donating or repurposing goods, and buying second-hand items are sustainable approaches.
- ‘Recycle’ promotes the reprocessing of materials to create new products, reducing the demand for new resources and lowering energy inputs used in manufacturing these goods.

Waste management techniques often rely on the individual doing the right thing, such as sorting their waste into recyclables and non-recyclables. However, in many countries this is not strictly enforced. People often feel disheartened with such schemes when large corporations often do very little to manage their waste which is produced on a much larger scale.

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Reducing greenhouse gas emissions and promoting alternative energy

Cutting emissions from human activities like burning fossil fuels for energy, deforestation, and industrial processes is a technique to slow down the pace of climate change. Greenhouse gas emissions, for example, dropped by 2% in 2022 across the European Union (EU), compared to 2021 levels, but accelerated action is needed to meet the EU’s ambitious climate and energy targets.

Alternative energy sources offer solutions to mitigate climate change impacts by reducing our dependence on fossil fuels. Alternatives to fossil fuels include renewable energy resources, such as solar, wind, wave, tidal, geothermal, biomass, and hydroelectric power. Each of these energy sources has the benefit of not emitting greenhouse gases and not being a finite resource, but they each have drawbacks. For example:

- solar and wind do not generate power all the time
- hydroelectric power disrupts ecosystems
- biomass can compete with food production
- geothermal power is limited to certain regions, usually along where tectonic plates meet.

Nuclear power is another alternative energy source that produces no greenhouse gases during production. It poses challenges, though, related to radioactive waste disposal and safety. You will learn more about the advantages and disadvantages of each of these energy sources in [Topic 10](#).



Figure 5.11: Svartsengi power station is a geothermal power station in Iceland

Afforestation and carbon capture

The UN Food and Agriculture Organisation estimates that tropical deforestation has contributed 8% of global carbon emissions. **Afforestation** reduces

greenhouse gases because trees absorb carbon dioxide from the atmosphere during photosynthesis. Afforestation is relatively cheap and effective. However, poorly managed projects can do more harm than good. For example, only planting one type of tree can reduce the biodiversity in that area, and some scientists argue that the amount of carbon that trees can store has been overestimated. It also takes a long time for trees to grow.

Carbon capture and storage (CCS) is the process of removing carbon dioxide from the waste gases from industrial processes. This carbon is then stored deep underground in aquifers or depleted oil and gas reservoirs. The main benefit of CCS is that it has the potential to reduce greenhouse gas emissions. However, CCS is expensive and some of the technologies are energy intensive. Some people worry that CCS poses an environmental risk from potential leakage, for example into groundwater sources, from long-term storage underground.

Electric cars

Electric cars can mitigate the impacts of climate change by reducing greenhouse gas emissions associated with traditional internal combustion engine vehicles. Electric cars powered by electricity from renewable sources produce fewer, if any, exhaust emissions. This shift to cleaner energy helps decrease the overall carbon footprint of the transportation sector. For instance, countries incentivising electric vehicle adoption, such as Norway, have seen a notable reduction in carbon dioxide emissions. Advancements in battery technology contribute to increased energy efficiency, cheaper battery costs, and the potential for widespread adoption of electric vehicles. As more regions invest in charging infrastructure and promote the transition to electric transportation, the net effect is a substantial decrease in air pollution and greenhouse gas emissions.

However, there are issues surrounding the production and disposal of the batteries used in electric vehicles. The extraction and processing of materials like lithium, cobalt, and nickel, which are essential components of lithium-ion batteries, can have environmental impacts. The energy-intensive manufacturing process of batteries contributes to carbon emissions, and the disposal and recycling of batteries poses environmental contamination issues. At present, manufacturers are not able to power vehicles like lorries and planes with batteries, as they would be too heavy with the number of batteries needed. Manufacturers are therefore looking to alternate technologies like hydrogen power for this.



Figure 5.12: Lithium production, shown here in Argentina, is needed for battery components in electric cars. The environmental impact of this can be high.

PROJECT

Rising sea levels are threatening this Pacific paradise

Good luck finding Tuvalu on a map. The Pacific paradise, located between Australia and Hawaii, is one of the smallest countries in the world. Its nine tiny islands make up just 10 square miles. It’s incredibly picturesque. But because of climate change, it’s also incredibly fragile. ‘The highest point in Tuvalu is only a few metres above sea level, so everybody is aware that if changes happen in the sea, this will immediately affect life there,’ said Sean Gallagher, a photographer and filmmaker who visited in March.

Tuvalu already struggles with the rising sea. Between January and March, its seasonal ‘king tides’ cause severe flooding. It’s been that way for decades. ‘Water literally seeps up through the ground below,’ Gallagher said. ‘This could happen in minutes, and I witnessed many communities being suddenly flooded. Even the airport runway was underwater at times. Locals told me that this is an increasing problem, and they worry about it getting worse as sea levels rise.’

The encroaching sea brings with it a host of problems. The country’s limited farmland can be poisoned by the salt water. Its groundwater is at risk. And the sea’s warming temperature presents a threat to the coral reefs that surround the islands and the fish that live there. ‘Climate change is having real-world impacts right now,’ Gallagher said. ‘It’s not a problem for the future. It’s a crisis for today.’

Source B: A 2019 CNN news report about the impacts of sea level rise on Tuvalu

Read Source B. Imagine your class are a team of environmental consultants hired by the government of Tuvalu to develop a sustainable management plan to mitigate and adapt to the impacts of climate change.

Your task is to assess the current situation, propose effective strategies, and create a management plan that considers the unique geographical, social, and economic context of Tuvalu. You will then present this as a class. You should conduct your research online and produce a slideshow to present.

The class could split into the following research groups:

Initial assessment

Research and evaluate the current impacts of climate change on Tuvalu, focusing on key indicators such as sea level rise, temperature change, and extreme weather events. Identify vulnerable areas, industries, and communities.

Stakeholder analysis

Identify key stakeholders involved in or affected by climate change in Tuvalu. You should include government agencies, local communities, non-governmental organisations (NGOs), and international organisations. Assess their interest in, influence, and potential contributions to the management plan.

Risk assessment

Conduct a comprehensive risk assessment considering both short-term and long-term impacts of climate change on Tuvalu. Rank the risks based on their severity and potential consequences.

Mitigation and adaptation strategies

Research and propose a range of mitigation and adaptation strategies that suit Tuvalu's specific needs. Rank the strategies based on their effectiveness and cost.

Adapting to the impacts of climate change

Earth's climate has changed a lot in the last one hundred years. Adapting to climate change involves adjusting to the actual or predicted future climate using the following techniques.

Coastal defences

Coastal defences can protect low-lying areas from sea-level rise and extreme weather events due to climate change. Countries like the Netherlands, for example, have built dikes, sea walls, and storm surge barriers to protect low-lying areas from flooding. 26% of the country currently lies below sea level, so these structures shield valuable agricultural land and critical infrastructure. They are, however, very expensive to construct. The Delta Works flood protection scheme in the Netherlands, for example, cost \$5 billion to construct and more to maintain and improve.



Figure 5.13: Part of the Delta Works flood protection scheme in the Netherlands, completed in 1997

Adapting homes

Homes need to be adapted to cope with a changing climate. Proper insulation helps regulate indoor temperatures, reducing the reliance on energy-intensive heating and cooling systems. Installing energy-efficient windows and using high-quality insulation materials such as fiberglass or foam can significantly decrease

heat transfer. This reduces the carbon footprint associated with excessive energy consumption. However, the costs of insulating a house can be very expensive and this puts the financial responsibility on individuals who want to reduce their environmental impact. Some governments provide grants to help towards the cost, but this often does not cover the whole cost. Furthermore, while many insulation materials are eco-friendly, some may have environmental consequences. Certain foam types, for example, may contain chemicals with environmental concerns and their production processes may have a higher carbon footprint than other more sustainable types.

Agriculture changes

Changing how we farm and grow crops that can withstand drought are crucial steps in adapting to climate change. For example, in Australia, where droughts are common, farmers are using smarter irrigation methods and harvesting rainwater to make the most of limited water. Similarly, in regions like sub-Saharan Africa and South Asia, farmers are planting crops that can handle drought conditions. This ensures a more stable food supply. For these changes to work, though, farmers need support like research, education, and financial help to make these practices widespread.

Water practices

With climate change leading to more frequent and intense weather events such as floods and droughts, enhanced water storage capabilities act as a buffer against water scarcity during dry periods. For example, large-scale engineering projects, like the Three Gorges Dam in China, help control floods and generate power, as well as managing water supplies. These kinds of projects require substantial investment, though, so may not be an easy solution for lower-income countries.

Improving drainage systems can help communities adapt to climate change. They mitigate the impact of heavy rainfall and flooding by channelling excess water away from urban areas. In the Netherlands, for example, a network of drainage channels protects against flooding.

Relocation

When the impacts of climate change become too severe, some communities, especially island nations, have no choice but to relocate. This means leaving their homes and starting over elsewhere, which is emotionally and financially challenging. Families lose not just their houses but also their connections to the land and their way of life. They become climate refugees. Moving entire communities is expensive and these costs often go beyond what affected countries can afford. This issue highlights the need for global efforts to address climate change.

Gardi Sugdub, or ‘Crab Island,’ is a tiny island in an archipelago off Panama’s northern Caribbean coast. It has been home to Guna Indigenous people for over 100 years. Originally the island provided the Guna with refuge from mainland mosquito-borne illnesses and colonial restrictions, but today, Gardi Sugdub faces new challenges.

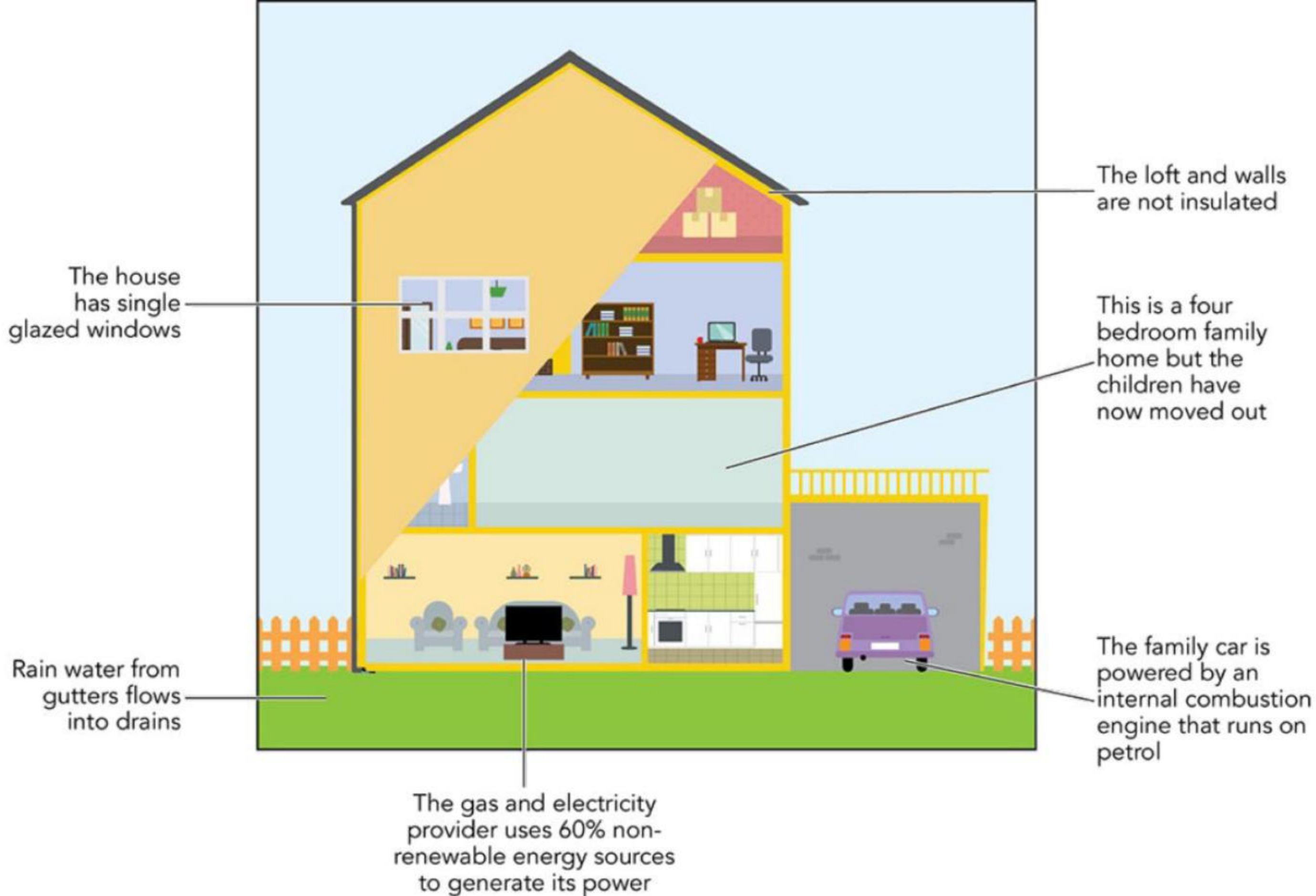
With nearly 1300 people on an island of about five soccer fields, there is no room to expand homes or for children to play. Floods and storms have made life even harder on the island, affecting housing, water, health, and education. Such extreme weather is only expected to become more common as the climate crisis accelerates. While urgent government action can still mitigate the worst climate change scenarios, scientists are also virtually certain of a rise in sea levels globally, undermining the habitability of the island these residents call home. Facing the inevitability of rising seas and surging storms on an island with no space, the Gardi Sugdub community have been contending with existential questions: what is next for our people? Do we have to move? If so, when, where, and how?

Human Rights Watch

Source : Relocating settlements and key infrastructure is a costly and socially disruptive solution in adapting to climate change

TASK 2: Study Source

- a** Describe the challenges faced by the Gardi Sugdub community.
- b** Propose potential government actions that could help mitigate the challenges faced by the inhabitants of Gardi Sugdub in the short-term and long-term.
- c** Evaluate the role of international co-operation and climate policies in addressing the challenges of small island communities like Gardi Sugdub. How might global initiatives impact the local resilience and adaptation strategies of such communities?



Source D: A conventional family home that is poorly designed and has a significant environmental impact

TASK 3: Study Source D

- What could be done to reduce the environmental impact of the family home in Source D?
- As a class, debate the success of individual actions, such as improving the environmental impact of family homes, versus implementing climate change policies.
- Evaluate how you contribute to climate change. Take a typical day, for example: how do you get around? How do you keep warm or cool? What do you eat? Are there any changes that you could make that could reduce your impact?

TASK 4

'It is better to mitigate the effects of climate change rather than adapt to them'. To what extent do you agree with this statement? Support your ideas with examples.